

Test aerosol system for calibration of automatic particulate dust measuring devices

Abstract

The system presented, the CCE 3000 (Calibration channel for environmental fine dust monitoring systems), allows production of test aerosols in the concentration range between 20 and 100 $\mu\text{g} / \text{m}^3$. The stability of the aerosol output is on average 6.9% over a test period of 72 hours. The metrological tracing is performed with the dust quantities applied, which are sampled with a small-filter device for 24 hours each and evaluated gravimetrically. For statistical confirmation, a DUT is tested thrice for 24 hours each, and the results are evaluated in the form of the daily mean values of the automatic process against the daily mean values of the small-filter process. Upon passing the test, an automatic measuring device, whose equivalence to the reference method was already confirmed under laboratory conditions, may thus enter the monitoring network operation. This is an important step in ensuring the data quality of continuously operating dust analysers.

1. Introduction

This paper presents a system for production of a stable test aerosol for the calibration of automatic particulate dust measuring devices (PM_{10} or $\text{PM}_{2.5}$). Up to now, automatic dust measuring devices have been tested for compliance with their zero value by placing a zero-air filter upstream into the sampling line. In addition, it is checked whether the nominal parameters of the device are within the tolerances according to the manufacturer's specification or according to own specifications. If the results of these tests are positive, the device is released for use. The extent to which the data supplied by the automatic measuring device are within the permissible equivalence corridor with respect to the reference method can be ascertained only afterwards by means of appropriate comparative measurements [1, 2, 3]. Direct metrological tracing is not possible. Typically, at least 6 weeks will elapse before it has been established whether the machine is operating correctly. This means that the procedures for automatic dust measuring devices are fundamentally different from those for most other analysers in monitoring network operation, where such a "blind flight" is not accepted.

As part of the modernisation and upgrading of the infrastructure for maintenance and quality assurance of dust measuring devices in the Saxon air monitoring network, test rigs for the determination of the dryer efficiency of FDMS modules and the production of test aerosols for the calibration of automatic dust measuring devices were implemented. In cooperation with the company Palas® Karlsruhe and the monitoring network operation Air of the BfUL in Saxony, the performance characteristics for the test aerosol system were developed. A prototype in various stages of development was installed and tested in the prepared low-dust and air-conditioned rooms. Technical details were modified over the course of about 2 years in order to match the performance of the facility to the given criteria. The maintenance and operating procedures also required optimisation in order to finally establish a practical solution. This has now been available since October 2016 and is described below.

2. Principle and technical structure

The following relevant framework conditions apply to the operation of the system:

Target concentration of particulate dust:	20 to 100 $\mu\text{g} / \text{m}^3$
Volume flow of the test devices:	1.0 m^3 / h (TEOM FDMS)
Maximum number of test devices:	2
Volume flow of the reference device:	2.3 m^3 / h (KF device Leckel)
Number of reference devices:	1
Maximum volume flow isokinetic sampling:	4.3 m^3 / h ($2 \times 1 \text{ m}^3 / \text{h} + 1 \times 2.3 \text{ m}^3 / \text{h}$)
Generation of the test dust:	Solids feeder RBG 1000
Main volume flow:	max. 20 m^3 / h
Dilution volume flow:	max. 20 m^3 / h
Test dust:	SAE fine (ISO 12103-1)

The maximum main and dilution volume flows are defined constructionally and by the available electrical power. With a higher main air volume flow, the particulate dust concentration that can be produced could be further reduced.

The RBG 1000 by Palas® is used to produce a stable test aerosol with very low dust concentrations [4, 5, 6]. In order to be able to calibrate the particulate dust measurement devices with the necessary dust concentration of approx. 20 – 100 $\mu\text{g} / \text{m}^3$, further measures have been taken on this test system. In order to adjust the particle size distribution approximately to the distribution of a PM₁₀ distribution, the test aerosol generated with the RBG 1000 is passed through a virtual impactor. At the same time, the aerosol concentration is diluted reliably and reproducibly by the virtual impactor. Thereafter, the particulate fraction from the virtual impactor is introduced into the test channel, diluted with the main volume stream to the desired concentration, and distributed homogeneously in the test channel. At the end of the test channel, the reference dust measurement device and a maximum of two particulate dust measuring devices (TEOM-1400 or TEOM-FDMS devices) to be calibrated are connected to the test channel via isokinetic sampling probes (see Fig. 2, in the basement). Depending on the total volume flow selected and the particulate dust measuring devices connected, appropriate, easily replaceable sampling probes are used.

The RBG system is schematically illustrated in Fig. 1.

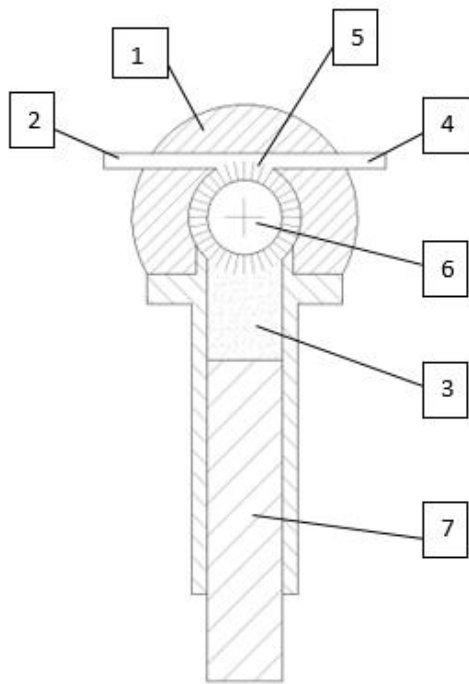


Figure 1 Brush metering device according to VDI 3491 sheet 3

- 1 Dispersing head
- 2 Dispersing gas inflow
- 3 Aggregate material / solid feed
- 4 Aerosol outlet
- 5 Point of origin of the disperse phase
- 6 Transport brush
- 7 Transport piston

The mass flow generated is calculated according to the following equation:

$$\dot{m}_p = \frac{m_p}{t} = \frac{v_p \cdot \rho_s}{t} = \frac{d^2 \cdot \pi}{4} \cdot w_k \cdot p_s$$

where:

- $\dot{m}_p$ Mass flow of particles
- m_p Mass of particles
- t Time
- v_p Volume of the particles in the solids container
- p_s Packing density of the powder in the solids container
- d Internal diameter of the solids container
- w_k Feed rate of the transport piston
- ρ_s Bulk density of the powder

The mass concentration is calculated according to the following equation:

$$c_m = \frac{\dot{m}_p}{\dot{v}}$$

where:

- c_m Mass concentration in kg / m³
- $\dot{m}_p$ Mass flow in kg / h
- $\dot{v}$ Volume flow in m³ / h

Three control parameters are available for adjusting the concentration of the test aerosol:

- Feed rate of the transport piston
- Main volume flow
- Dilution volume flow

Mass Flow Controllers (MFCs) are used for the exact and reproducible adjustment of the three volume flows of dispersion volume flow at the RBG, dilution volume flow at the virtual impactor, and main volume flow in the test channel.

Since the measurements are traced to the dust mass determined using a small-filter device as reference method according to **DIN EN 12341**, one test cycle takes 24 hours. Here at a dust concentration of $50 \mu\text{g} / \text{m}^3$ and a volume flow of the reference measuring device of $2.3 \text{ m}^3 / \text{h}$, a weighable dust volume of 2.7 mg per filter accrues.

The schematic illustration in Fig. 2 shows the structure of the system, which extends over two floors. Dust metering and production of the test aerosols take place on the upper floor. In a vertical tubular channel, the aerosol is then led to the floor below, where it is isokinetically removed for testing. Most of the test aerosol is led outward through a subsoil bypass line and filtered.

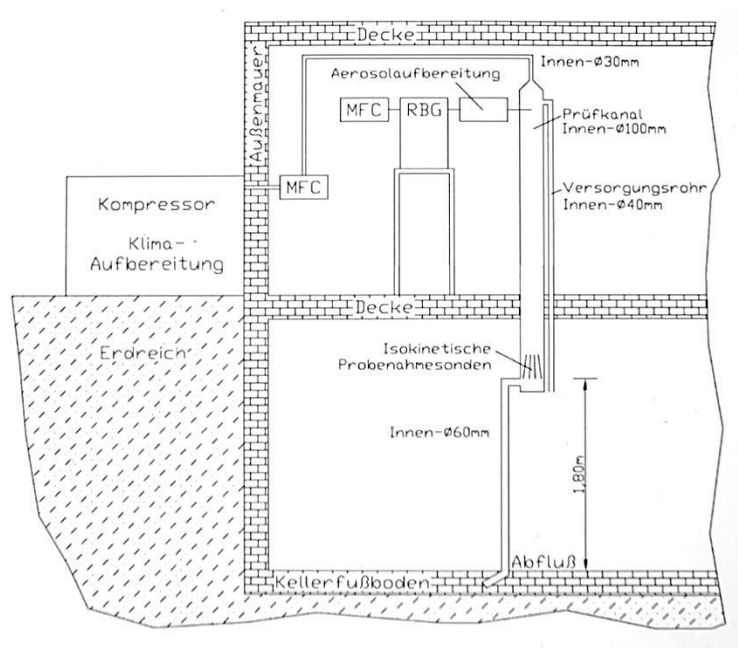


Fig. 2: Schematic structure of the system



Fig. 3: RBG 1000 with aerosol channel

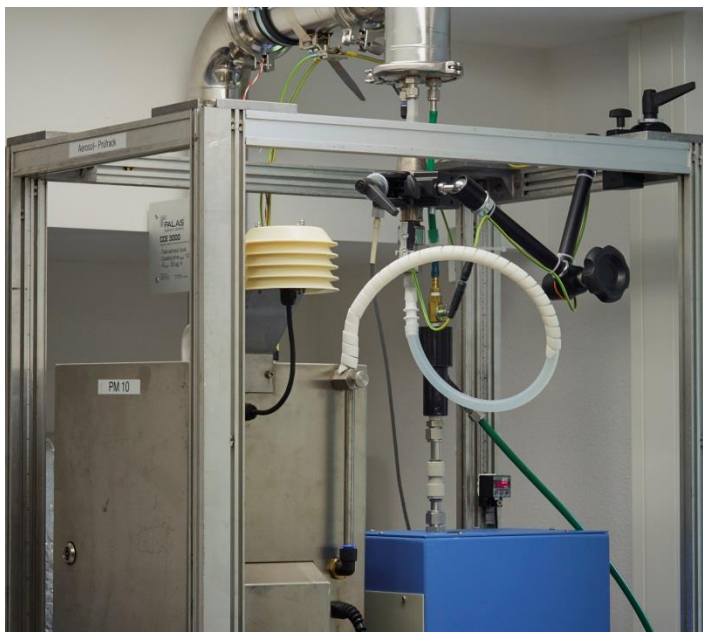


Fig. 4: Isokinetic sampling, reference and test device



Fig. 5 Test rig with small-filter unit and TEOM 1405



Fig. 6: Detail of isokinetic sample taking



Fig. 7: Test rig with zero-air compressor

3. Metrological tracing

The metrological tracing of the test is based on gravimetric determination of the particulate dust according to DIN EN 12341. A test interval of 24 hours has proven optimal – on the one hand for the production of the test aerosol within the limits of the specification (20 to 100 $\mu\text{g} / \text{m}^3$), the loading of the filters with a dust mass of low measurement uncertainty, and the service life of the dust briquette in the RBG 1000. On the other hand, in the form of the dust mass a daily mean value over 24 hours is available for this period. The extent to which the output of the test aerosol was stable in the cycle should first be determined by means of an additional, automatic dust measurement device. It has been shown that this additional effort is not required, because the stability test can be performed just as well with the device being tested. This measure has led to a marked streamlining of the test set-up.

With the described approach, it can be ensured that the proffered total mass of the test dust is determined for the test cycle. By monitoring the volumetric flow of the reference devices with test equipment traceable to the national reference, the average dust concentration can additionally be determined. The metrological tracing of the tested, automatic dust analysers can thus be carried out via their 24-hour average, including the determined standard deviation. In addition, the short-term values of the devices provide sufficient information on the stability of the test aerosol and reliably detect disturbances or failures of aerosol generation during the test cycle.

4. Results

Several parameters to assess the operability and usability of the test rig have been gradually identified. The most critical parameter is the stability of the aerosol yield. Initial tests showed low stability, which as a matter of principle manifested in tendencies to decreases as well as increases in dust concentrations, additionally overlaid by individual events, in which the values rose abruptly and briefly. All the correction methods used have focused on avoidance of deposits of the test dust in the system by improving the quality of the surfaces used, dissipating static charges of aerosol-contacting parts in the sampling as well, and establishing a regular maintenance phase. During the latter, the system is opened and cleaned. Operating conditions and maintenance costs are the resulting determining factors for the usable test capacity of the plant.

4.1. Stability of the aerosol concentration

In order to assess the stability of the aerosol yield, the measured values of the automatic dust measurement device to be tested are used. Here it is irrelevant whether the absolute dust measurement value of the device is correct. On the basis of the routine checks carried out since October 2016, data have been collected that can be used to assess stability. The results are shown in the form of the calculated arithmetic mean values and the standard deviations as a table and graphically as the course of dust concentration over time in Table 1 and in Figures 8 to 10, respectively. Generally, the test cycle is divided into a section during which dust is metered continuously for two to three days, and a part during which the dust concentration is determined without metering (zero value of the system). The total cycle can take up to 5 days.

Table 1: Mean values and standard deviations for test aerosol yields and zero values

Test cycle	Type	1 st day	2 nd day	3 rd day	Total cycle
1	P	47.5 $\mu\text{g}/\text{m}^3 \pm 4.2$	51.1 $\mu\text{g}/\text{m}^3 \pm 3.2$	63.0 $\mu\text{g}/\text{m}^3 \pm 4.3$	53.9 $\mu\text{g}/\text{m}^3 \pm 8.2$
2	P	75.3 $\mu\text{g}/\text{m}^3 \pm 6.1$	66.0 $\mu\text{g}/\text{m}^3 \pm 3.2$	70.9 $\mu\text{g}/\text{m}^3 \pm 4.3$	70.7 $\mu\text{g}/\text{m}^3 \pm 4.6$

3	P	$81.1 \mu\text{g}/\text{m}^3 \pm 7.2$	$79.9 \mu\text{g}/\text{m}^3 \pm 3.8$	$87.8 \mu\text{g}/\text{m}^3 \pm 4.9$	$82.9 \mu\text{g}/\text{m}^3 \pm 4.3$
4	P	$61.1 \mu\text{g}/\text{m}^3 \pm 2.5$	$61.9 \mu\text{g}/\text{m}^3 \pm 3.4$	$68.4 \mu\text{g}/\text{m}^3 \pm 4.2$	$63.8 \mu\text{g}/\text{m}^3 \pm 4.0$
5	P	$34.5 \mu\text{g}/\text{m}^3 \pm 1.9$	$36.0 \mu\text{g}/\text{m}^3 \pm 2.9$	$39.2 \mu\text{g}/\text{m}^3 \pm 2.3$	$36.6 \mu\text{g}/\text{m}^3 \pm 2.4$
6	P	$43.2 \mu\text{g}/\text{m}^3 \pm 12.9$	$38.4 \mu\text{g}/\text{m}^3 \pm 3.3$	$42.3 \mu\text{g}/\text{m}^3 \pm 3.0$	$41.3 \mu\text{g}/\text{m}^3 \pm 2.5$
1 – 6	N				$3.5 \mu\text{g}/\text{m}^3 \pm 1.5$

P = test with dust; N = zero value of the system without dust metering

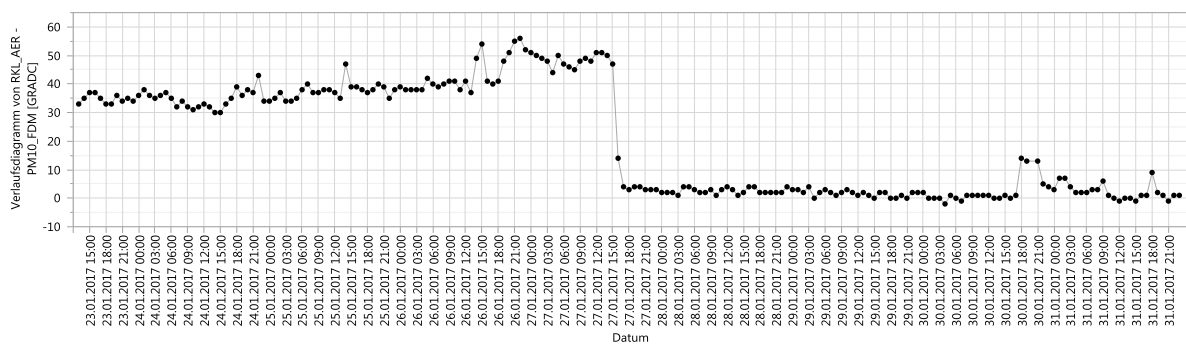


Fig. 8: Course of the dust concentration over the entire cycle

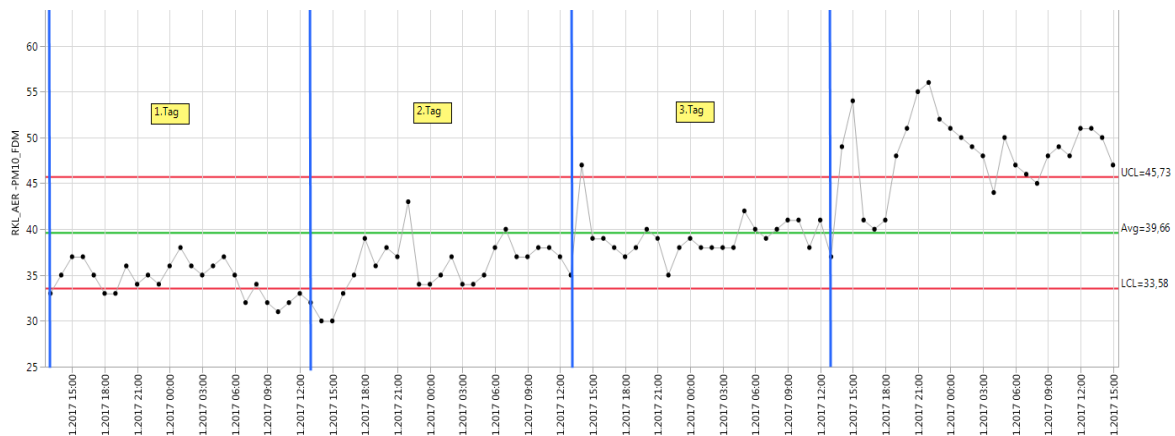


Fig. 9: Course of the test aerosol concentration in the measuring cycle

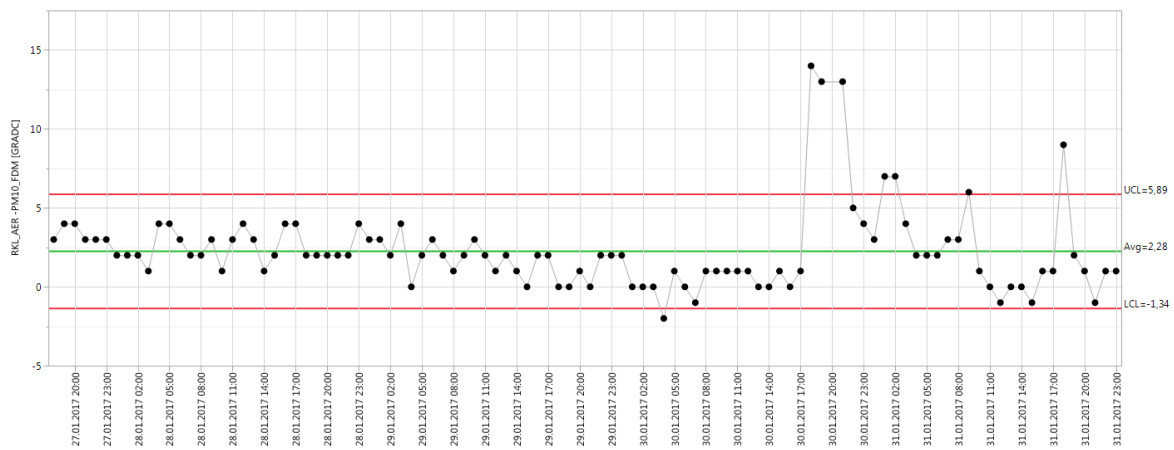


Fig. 10: Course of the system zero value

4.2. Contaminations

Contaminations in connection with the operation of the test aerosol system are stray amounts of dust that are generated in the system due to settling. Fig. 10 shows such an effect. It is characterised by values increasing relative to the rest of the course, which can occur, as the metering is deactivated, only due to dust in the system that has not been dispersed yet. In the course of system and process optimisation, the effects have been reduced to an acceptable minimum. As shown in Table 1, the system provides zero values on the order of $3.5 \mu\text{g} / \text{m}^3$ on average, with a corresponding standard deviation of $1.5 \mu\text{g} / \text{m}^3$.

As a matter of principle, with the test aerosol system it is also possible to examine dust not only in terms of its concentration but also of its constituents. Such questions have not been addressed so far, so that contamination in the sense of foreign entry of substances that are not originally contained in the test dust or enter in significantly increased amounts cannot be assessed.

4.3. Protocol for TEOM FDMS

A protocol has been developed for the functional testing of the dust machines that documents the test conditions and results and includes the approval of the measuring instrument for use in the monitoring network. As an example, such a protocol is shown in Fig. 11.

Prüfprotokoll Staub								
Probenahme mit TEOM FDMS +Kleinfilter								
Messkomponente:	Staub PM 10					Datum:	16.01.2017	
Anlage:	Kalibriergasanlage Staub					Uhrzeit Start:	13:00	[MEZ]
Prüfling:						Uhrzeit Stopp:	13:00	[MEZ]
	l (mm)	d (mm)	A (mm ²)	A (m ²)	m3/h	V1 (m/h)	V1 (m/s)	
Kanalquerschnitt		81	5150,4	0,00515		2831,07	0,79	
Teilquerschnitt		18	254,3	0,00025				
3 X Teilquerschnitt			763,02	0,0007630		19109,7	5,31	
Düse_1	30	7,9	49,0	0,0000490	1	20411,6	5,67	
Düse_2	30	10	78,5	0,0000785	1,6	20382,2	5,66	
Düse_3	30	7,9	49,0	0,0000490	1	20411,6	5,67	
mit einem FDMS		-D1, -D2		0,0006355		22943,3	6,37	
mit FDMS + Kleinfilter		nur -D1		0,0007140		20420,9	5,67	
Vorlauf nur FDMS:								
Vol. Strom_1	Verd. Strom	Gesamtstrom	gesamt		t _{Aer}	p _{Atm}	Vol.Str. RBG	Vorsch. Gesch.
NI/min	NI/min	NI/min	Nm ³ /h	Bm ³ /h	°C	hPa	NI/min	mm/h
192	30	222	13,32	14,58	19	990	42	2
Konzentration TEOM FDMS (µg/m ³)	34,48					Kompr. WS (h)	4747,60	
Konzentration Kleinfiltergerät (µg/m ³)	36,48					Kompr. WS (h)	4765,80	75,8%
						Kompr. groß (h)	1944,50	
	Nm ³	m ³	h			Kompr. groß (h)	1953,20	36,3%
Kleinfiltergerät:	35,51	38,38	24			Restvorsch. (mm)	62,7	
m (Kleinfilter) in g:	0,0014					Restvorsch. (mm)	43,5	0,80
Bearbeiter:	M. Eilzer					Filter #		50972
Unterschrift:								
Vorschub RBG = 2.0mm								

Fig. 11: Test protocol RKL-AER

4.4. Maintenance effort

- After each sampling with small-filter device: Cleaning of the filter holder with compressed air
- After one test run (corresponds to 3 to 4x24 h sampling of test aerosol plus approx. 8 h stabilisation-time) = total between 80 h and approx. 100 h of operating time:
 - Removal and cleaning of the feed reservoir for dust on the RBG 1000 (by brush + vacuum cleaner)
 - Removal and cleaning of the virtual impactor (brush + vacuum cleaner)
 - Removal and cleaning of the connecting element between the RBG1000 and the virtual impactor (brush + vacuum cleaner)
 - Control of the compressor run times, control of the dryer units of the compressors (dew point, remaining time to service), draining of the condensate tank
- After approx. 5 to 6 measuring cycles (approx. 500 h) or before commissioning after a longer downtime:
 - See above, additionally cleaning with compressed air
 - Removal and cleaning of the sampler (compressed air)
 - Cleaning of the exhaust hose of the RBG1000
 - Visual inspection of the oil separator
 - Removal and visual inspection of the brush on the RBG1000

4.5. Testing capacity

The currently available testing capacity is sufficient for one automatic dust measurement device per week, comprising at least two complete 24-hour tests. Approximately 25 of these devices, which are serviced annually, are being used in the Saxon air monitoring network. This means that the annual volume of test items is 25 units, which can be easily tested in 52 calendar weeks. Depending on the device

system to be tested, doubling of the capacity is possible if the second connection on the isokinetic sampling device is also used.

5. Outlook

Using the aerosol test rig in the form described, further stabilisation of the data quality for the automatic dust measuring devices used in Saxony will be achieved, in particular thanks to the possibility of checking devices for correct operation by means of a function check immediately before the monitoring network application, whereby deviations from the equivalence corridor, so far detectable only after several weeks of operation, can be avoided. Another field of application is the use of the test aerosol system for determination of the contents of dust. In particular, determination of elementary carbon is to be named as an issue of special interest, since there is currently no base reference method for this measurand. In function control with soot-comprising dust, both the behaviour of automatic measuring instruments as well as the laboratory analysis of elementary carbon with filter material and the currently used thermographic and thermal-optical analytical methods is of interest.

6. References

- /1/ Bruns, Klaus, Wie sichert man die Qualität einer Qualitätssicherung? HLUG Jahresbericht 2007
- /2/ Nachweis der Äquivalenz für automatische PM10-Messungen mit dem Staubmessgerät SHARP 5030, Staatliches Gewerbeaufsichtsamt Hildesheim, 2014
- /3/ Kath, Horst-Günter, Vergleichsmessungen zur Bestimmung der Feinstaubkonzentration für PM2.5 im städtischen Hintergrund Dresdens, Betriebsgesellschaft für Umwelt und Landwirtschaft, Radebeul, 2014
- /4/ Zahradnicek, A, Methoden zur Aerosolherstellung aus vorgegebenen Haufwerken, Staub – Reinhaltung der Luft 35/6 (1975)
- /5/ Löffler, F., Zahradnicek, A., Eine neue Dosiervorrichtung zur Erzeugung von Aerosolen aus vorgegebenen feinkörnigen Feststoffen , Staub – Reinhaltung der Luft 36/11 (1976)
- /6/ VDI Richtlinie 3491 Blatt 3 „Messen von Partikeln – Herstellungsverfahren für Prüfaerosole – Dispergierung von Haufwerken und Feststoffen“

Authors: Klemens Regir, Horst-Günter Kath¹

 Leander Mölter²

¹ Betriebsgesellschaft für Umwelt und Landwirtschaft, Messnetzbetrieb Luft
Altwahnsdorf 12, D-01445 Radebeul

² Palas® GmbH, Greschbachstraße 3b, D-76229 Karlsruhe